

</> Python

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier

from xgboost import XGBClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_auc_score

import joblib
```

Model building is the core phase of the Flood Prediction System, where machine learning algorithms learn the relationship between meteorological parameters and flood occurrences. The primary objective is to develop a classification model capable of predicting whether a flood is likely to occur based on historical rainfall patterns.

In this project, four supervised machine learning algorithms were implemented and compared:

- Decision Tree
- Random Forest
- K-Nearest Neighbours (KNN)
- Extreme Gradient Boosting (XGBoost)

Each model was trained using historical rainfall data and evaluated using standard classification metrics. The best-performing model was selected for deployment in the Flask web application.

Feature	Description
ANNUAL	Total annual rainfall
Jan-Feb	Rainfall during January–February
Mar-May	Rainfall during March–May
Jun-Sep	Rainfall during June–September
Oct-Dec	Rainfall during October–December
YEAR	Observation year